

TOC2

Probability and Statistics I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A real-estate analyst studies 412 home sales in a neighborhood. Most sold for \$200,000 to \$300,000, but a few estates sold for more than \$2 million. The data came from the county's public records office. Which statement about the mean and median sale price is most likely true?

- A. They are about equal because 412 sales is a large sample.
- B. The median always equals the mean when data come from public records.
- C. The mean is less than the median.
- D. The mean is greater than the median.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher's class of 31 took a quiz on a Friday. Scores had a mean of 72 and a standard deviation of 8. Because one question was flawed, she adds 5 points to every score. What are the new mean and standard deviation?

- A. Mean 77, SD 8
- B. Mean 77, SD 3
- C. Mean 77, SD 13
- D. Mean 72, SD 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A data set has a mean of 40 and a standard deviation of 6. Every value is multiplied by 1.5. What are the new mean and standard deviation?

- A. Mean 60, SD 6
- B. Mean 41.5, SD 7.5
- C. Mean 60, SD 9
- D. Mean 60, SD 13.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A company surveyed 200 employees about commute time after it moved offices in March. The box plot shows minimum 5, Q1 15, median 22, Q3 40, and maximum 70 minutes. About how many employees commute between 15 and 40 minutes?

- A. About 50
- B. About 100
- C. About 25
- D. About 150

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A sleep researcher collected data from 150 volunteers through a phone app. The correlation between daily screen time and hours of sleep is $r = -0.85$. The app is free to download. Which statement is correct?

- A. About 85% of the variation in sleep is explained by screen time.
- B. The relationship is weak because r is negative.
- C. Screen time causes about 72% of lost sleep.
- D. About 72% of the variation in sleep is explained by the linear relationship with screen time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A course with 9 sections uses the regression line $\hat{y} = 40 + 6x$ to predict exam score from hours studied. One student studied 7 hours and scored 78. The course coordinator plans to share the model at an August workshop. What is this student's residual?

- A. +4
- B. 82
- C. 38
- D. -4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

For the line $\hat{y} = 20 + 4x$, a data point has an actual y of 70 and a residual of -6 . What is its x -value?

- A.** 14
- B.** 12.5
- C.** 76
- D.** 11

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A town council posts a poll on its website asking whether to build a skate park. The poll stays open for two weeks and gets 2,300 responses. What is the main problem with using the results to describe all residents?

- A.** None; 2,300 responses is large enough to remove bias.
- B.** Voluntary response bias: people with strong opinions are more likely to respond.
- C.** Convenience sampling by the council.
- D.** Response bias from leading question wording.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A district tracked 600 students who chose whether to attend after-school tutoring. The tutored group scored higher on the spring test. The program was funded by a local foundation. Can the district conclude that tutoring caused the higher scores?

- A.** No; students who chose tutoring may differ, for example in motivation, so random assignment is needed.
- B.** No, because observational studies can never show any relationship.
- C.** Yes, because 600 students is a large sample.
- D.** Yes, if the difference is statistically significant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A researcher wants evidence that a new reading app causes gains in reading scores. Which design supports that claim?

- A.** Compare two classes whose teachers chose whether to use the app.
- B.** Randomly sample students and compare those who already use the app with those who do not.
- C.** Randomly assign volunteers to use the app or not, then compare gains.
- D.** Survey app users about how much they improved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A café that opened in 2019 finds that 40% of customers buy coffee, 30% buy a pastry, and 18% buy both. The owner has two locations. What is the probability that a customer buys coffee or a pastry?

- A.** 0.70
- B.** 0.52
- C.** 0.48
- D.** 0.34

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

$P(A) = 0.5$, $P(B) = 0.5$, and $P(A \text{ or } B) = 0.8$. What is $P(A \text{ and } B)$?

- A.** 1.0
- B.** 0.25
- C.** 0.2
- D.** 0.3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A real-estate analyst studies ~~412 home sales~~ in a neighborhood. Most sold for \$200,000 to \$300,000, but a few estates sold for more than \$2 million. ~~The data came from the county's public records office.~~ Which statement about the mean and median sale price is most likely true?

Target word: mean vs. median **Decides:** A few very high prices create a right skew.

A	Camouflage	Sample size does not remove skew.
B	Absolute language	The data source does not decide the shape.
C	Inverted operation	That pattern comes from a left tail of unusually low values.
D	Correct	The extreme high values pull the mean up; the median stays near the typical sale.

2. Answer A Concept 2

A teacher's ~~class of 31 took a quiz on a Friday.~~ Scores had a mean of 72 and a standard deviation of 8. Because one question was flawed, she adds 5 points to every score. What are the new mean and standard deviation?

Target word: effect of adding a constant **Decides:** Adding shifts center, not spread.

A	Correct	Every score moves up 5; distances between scores do not change.
B	Inverted operation	Subtracts 5 from the SD.
C	Adjacent category	Adds 5 to the spread as well as the center.
D	Dropped element	Leaves out the shift in the mean.

3. Answer C Concept 2 · transfer

A data set has a mean of 40 and a standard deviation of 6. Every value is multiplied by 1.5. What are the new mean and standard deviation?

Target word: effect of multiplying by a constant **Decides:** Scaling multiplies both center and spread.

A	Dropped element	Applies the adding rule; multiplying does change spread.
B	Inverted operation	Adds 1.5 instead of multiplying.
C	Correct	$40 \times 1.5 = 60$ and $6 \times 1.5 = 9$.
D	Confusion set	Multiplies by $1.5^2 = 2.25$, the factor for variance, not SD.

4. Answer B Concept 3

A company surveyed 200 employees about commute time ~~after it moved offices in March.~~ The box plot shows minimum 5, Q1 15, median 22, Q3 40, and maximum 70 minutes. About how many employees commute between 15 and 40 minutes?

Target word: count in the box **Decides:** Q1 to Q3 holds the middle 50%.

A	Confusion set	Reads the 50% as a count.
B	Correct	$50\% \text{ of } 200 = 100$.
C	Wrong stem number	$40 - 15 = 25$ is the IQR in minutes, not a count.
D	Adjacent category	That is 75%, the share below Q3.

5. Answer D Concept 4

A sleep researcher collected data from 150 volunteers through a phone app. The correlation between daily screen time and hours of sleep is $r = -0.85$. The app is free to download. Which statement is correct?

Target word: interpret r and r^2 **Decides:** $r^2 = (-0.85)^2 \approx 0.72$.

A	Dropped element	Uses r without squaring it.
B	Confusion set	The sign gives direction; $ r = 0.85$ is strong.
C	Adjacent category	Correlation from observational data does not show cause.
D	Correct	$r^2 = 0.7225$.

6. Answer D Concept 5

A course with 9 sections uses the regression line $\hat{y} = 40 + 6x$ to predict exam score from hours studied. One student studied 7 hours and scored 78. The course coordinator plans to share the model at an August workshop. What is this student's residual?

Target word: residual **Decides:** $\hat{y} = 40 + 6(7) = 82$; residual = $78 - 82$.

A	Inverted operation	Subtracts observed from predicted.
B	Confusion set	That is the predicted score.
C	Dropped element	$78 - 40$ drops the slope term.
D	Correct	Observed 78 minus predicted 82.

7. Answer A Concept 5 • transfer

For the line $\hat{y} = 20 + 4x$, a data point has an actual y of 70 and a residual of -6 . What is its x -value?

Target word: solve for x from a residual **Decides:** Residual = $y - \hat{y}$, so $\hat{y} = 70 + 6 = 76$.

A	Correct	$20 + 4x = 76$ gives $x = 14$.
B	Dropped element	Drops the residual and solves $20 + 4x = 70$.
C	Confusion set	That is the predicted value, not x .
D	Inverted operation	Uses $\hat{y} = 70 - 6 = 64$.

8. Answer B Concept 6

A town council posts a poll on its website asking whether to build a skate park. The poll stays open for two weeks and gets 2,300 responses. What is the main problem with using the results to describe all residents?

Target word: type of bias **Decides:** People choose whether to respond.

A	Absolute language	Size does not fix a biased method.
B	Correct	Self-selected samples overrepresent strong views.
C	Confusion set	Close, but the respondents select themselves; that is voluntary response.
D	Adjacent category	Nothing in the stem suggests loaded wording.

9. Answer A Concept 7

A district tracked 600 students who chose whether to attend after-school tutoring. The tutored group scored higher on the spring test. ~~The program was funded by a local foundation.~~ Can the district conclude that tutoring caused the higher scores?

Target word: causal claim from observational data **Decides:** Students chose their own group.

A	Correct	Self-selection creates possible confounding.
B	Absolute language	They can show association, just not cause.
C	Wrong stem number	Sample size does not remove confounding.
D	Adjacent category	Significance shows a real difference, not its cause.

10. Answer C Concept 7 • transfer

A researcher wants evidence that a new reading app causes gains in reading scores. Which design supports that claim?

Target word: design that supports causation **Decides:** Random assignment to treatment.

A	Dropped element	Drops random assignment; teacher choice can confound.
B	Confusion set	Random sampling is not random assignment; users chose the app.
C	Correct	Random assignment balances confounders.
D	Adjacent category	Self-reports with no comparison group.

11. Answer B Concept 8

A café ~~that opened in 2019~~ finds that 40% of customers buy coffee, 30% buy a pastry, and 18% buy both. ~~The owner has two locations.~~ What is the probability that a customer buys coffee or a pastry?

Target word: P(A or B) **Decides:** Subtract the overlap: $0.40 + 0.30 - 0.18$.

A	Dropped element	Counts the 18% who buy both twice.
B	Correct	$0.70 - 0.18 = 0.52$.
C	Confusion set	That is P(neither), the complement.
D	Adjacent category	That is exactly one item, not at least one.

12. Answer C Concept 8 • transfer

$P(A) = 0.5$, $P(B) = 0.5$, and $P(A \text{ or } B) = 0.8$. What is $P(A \text{ and } B)$?

Target word: solve for the overlap **Decides:** $0.8 = 0.5 + 0.5 - P(A \text{ and } B)$.

A	Dropped element	Adds $P(A)$ and $P(B)$ and ignores $P(A \text{ or } B)$.
B	Confusion set	Assumes independence (0.5×0.5); the stem does not say so.
C	Correct	$1.0 - 0.8 = 0.2$.
D	Inverted operation	Subtracts one probability from the union: $0.8 - 0.5$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mean vs. median under skew	A long right tail pulls the mean above the median; the median resists extreme values.	Thinking a large sample forces mean and median to match.	A few extreme values in one direction.
2	Linear transformations of data	Adding a constant shifts the mean but not the standard deviation; multiplying by a constant scales both.	Adding the constant to the standard deviation, or leaving the SD unchanged when scaling.	“Every score is increased by” or “multiplied by.”
3	Reading a box plot	Each of the four sections of a box plot holds about 25% of the data; the box holds the middle 50%.	Reading a section’s length as its count.	A five-number summary and a question about how many values.
4	Correlation and r^2	r measures linear strength and direction; r^2 is the share of variation in y explained by the linear model.	Using r as the percent explained, or reading correlation as cause.	A value of r and “what percent of the variation.”
5	Predictions and residuals	Residual = observed y – predicted $\hat{y}$; negative means the point lies below the line.	Subtracting in the wrong order, or reporting $\hat{y}$ as the residual.	A regression equation and one actual data point.
6	Sampling bias	Voluntary response samples overrepresent people with strong opinions; a large biased sample is still biased.	Believing a big sample removes bias.	An online or call-in poll with many responses.
7	Experiments vs. observational studies	Only random assignment to treatments supports cause-and-effect conclusions; observational studies can be confounded.	Treating random sampling, or a large sample, as proof of causation.	Groups that chose their own treatment.
8	The addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.	Forgetting to subtract the overlap, or assuming independence.	Two overlapping percentages and the word “or.”

Course scope (public description): Probability and Statistics I covers the knowledge and skills necessary to apply basic probability, descriptive statistics, and statistical reasoning and to use appropriate technology to model and solve real-life problems. It provides an introduction to the science of collecting, processing, analyzing, and interpreting data, including representations, constructions, and interpretation of graphical displays (e.g., box plots, histograms, cumulative frequency plots, scatter plots). Topics include creating and interpreting numerical summaries and visual displays of data; regression lines and correlation; evaluating sampling methods and their effect on possible conclusions; designing observational studies, controlled experiments, and surveys; and determining probabilities using simulations, diagrams, and probability rules. College Algebra is a prerequisite to this course.